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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 200,470,80,150,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **2**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Sorting the entire list sequentially
- b. Combining sorted sublists
- c. Comparing each element with the others
- d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **3**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Duplicates
- b. All of the above
- c. Selection
- d. Searching

The correct answer is: All of the above

Question **4**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. order()
- c. arrange()
- d. sort()

The correct answer is: sorted()

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It compares adjacent elements and swaps them if necessary
- b. It repeatedly finds the minimum element and moves it to the sorted part of the list
- c. It builds a sorted array one element at a time
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **6**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Using the divide-and-conquer approach
b. Being the most efficient sorting algorithm
c. Bubbling up the largest element to its correct position with each pass
d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Merge Sort
b. Binary Search
c. Bubble Sort
d. Linear Search

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Bubble Sort
c. Merge Sort
d. Linear Search

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It simplifies the sorting process by using only one pass
c. It allows for efficient parallel processing and sorting of data
d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **10**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is based on the divide-and-conquer approach
b. It is less efficient than bubble sort
c. It sorts data using a single pass
d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Merge Sort
c. Binary Search
d. Linear Search

The correct answer is: Merge Sort

Question **12**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of all keys

b. A sorting algorithm is stable if it preserves the order of duplicate keys

c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort

b. Merge Sort

c. Bubble Sort

d. Insertion Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging

b. Complexity

c. Searching

d. Sorting

The correct answer is:
Complexity

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting helps in finding duplicates quickly
- b. Sorting makes it harder to search for items
- c. Sorting decreases the efficiency of selection operations
- d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly